GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

# Solving Problems with Rational Numbers

*Form A · 29 points*

Name_____
Date_____
Period_____

*Show your work. One point for each lettered part.*

## Part 1. Operations in Scientific Notation

*12 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** Add or subtract. Write each answer in scientific notation.

**a.** $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

**b.** $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

**c.** $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

**d.** $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

**2.** Multiply or divide. Write each answer in scientific notation.

**a.** $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

**b.** $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

**c.** $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

**d.** $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

**3.** Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$ ? *1 pt*

- ○ A. $3.04 \times 10^5$
- ○ B. $4.3 \times 10^5$
- ○ C. $2.914 \times 10^5$
- ○ D. $4.3 \times 10^8$

**4.** Select ALL of the expressions whose value is $3 \times 10^6$ . *1 pt*

- ☐ A. $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☐ C. $(6 \times 10^3)(5 \times 10^2)$
- ☐ D. $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☐ F. $\frac{9 \times 10^9}{3 \times 10^3}$

**5.** Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$ . Name the two mistakes she made, and give the correct sum. *1 pt*

**6.** A number is multiplied by $4 \times 10^3$ and the product is $2 \times 10^{-1}$ . What was the number? Write your answer in scientific notation. *1 pt*

## Part 2. Real-World Problems in Scientific Notation *4 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**7.** Solve. Write each answer in scientific notation. Do not round.

**a.** A rectangular plate measures $2.4 \times 10^{2}$ mm by $1.35 \times 10^{2}$ mm. Find its area. *1 pt*

**b.** A vehicle travels $5.6 \times 10^{2}$ km in 8 hours. Find its average speed. *1 pt*

**8.** Solve. Write each answer in scientific notation. Do not round.

**a.** Two lengths are measured: $6.25 \times 10^{3}$ mm and $4.1 \times 10^{2}$ mm. Find their sum. *1 pt*

**b.** Two masses are measured: $9.3 \times 10^{-2}$ g and $4.72 \times 10^{-3}$ g. Find their difference. *1 pt*

## Part 3. The Order of Operations with Exponents and Radicals *13 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** Evaluate each expression.

**a.** $15 - 2 \cdot 3^2$ *1 pt*

**b.** $\frac{5^2 + 7}{2^3}$ *1 pt*

**c.** $36 \div 4 \cdot 3$ *1 pt*

**d.** $-4^2 + (-4)^2$ *1 pt*

**10.** Evaluate each expression.

**a.** $\sqrt{6^2 + 13}$ *1 pt*

**b.** $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

**c.** $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

**11.** A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

**a.** Write an expression for the total cost. *1 pt*

**b.** Find the total cost. *1 pt*

**12.** Which is the value of $2 + 5 \cdot 3^2$ ? *1 pt*

- ○ A. 32
- ○ B. 47
- ○ C. 63
- ○ D. 289

**13.** Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☐ B. $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☐ D. $\left(\frac{1}{6}\right)^{-1}$
- ☐ E. $\sqrt{5^2 + 11}$
- ☐ F. $\frac{4^2 + 8}{4}$

**14.** Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$ . Explain why she is wrong, and give the correct value. *1 pt*

**15.** A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*